ENEL700 Tutorial 4

1. An 162MHz carrier is deviated by 12kHz by a 2kHz modulating signal. What is the modulation index? Is the resulting FM signal a narrow band FM signal?

2. For a single frequency sine wave modulating signal of 3kHz with a carrier frequency of 36MHz, what is the spacing between sidebands? Calculate the bandwidth occupied by the signal using Carson's rule if the frequency deviation is 12kHz.

3. For a FM signal with frequency deviation of 8kHz and its modulating signal with frequency of 2kHz, what is the minimum bandwidth of the signal that contains 99% of its signal power?

4. A 5kHz sinusoidal signal with amplitude 5V serves as the modulating signal for a FM modulator of sensitivity of 15kHz/V, determine the Carson's bandwith of the resulting FM signal.

5. A carrier of 49MHz is frequency-modulated by a 1.5kHz square wave. The modulation index is 0.25. Sketch the spectrum of the resulting signal, assuming that only harmonics less than the sixth are passed by the system.